STANDATA interpretation 23-FCI-010/23-BCI-043

Fire/Building

Electromagnetic locks – installation, maintenance, testing, and verification

Date Issued: November 2025

Purpose

To clarify how the *Safety Codes Act (SCA)*, the National Building Code – 2023 Alberta Edition (NBC(AE)) and the National Fire Code – 2023 Alberta Edition (NFC(AE)) apply to the installation, maintenance, testing, and verification of electromagnetic locks.

Discussion

Electromagnetic locks (maglocks) use magnetic force to secure doors and are typically integrated with access control systems such as keypads, card readers, or remote release mechanisms. When energized, the magnet generates a strong holding force that keeps the door locked, making maglocks a common choice for security applications. However, maglocks must be integrated in a way that does not compromise fire protection systems or obstruct exit routes. Maglocks must be fail-safe, releasing upon fire alarm activation, power loss, or manual override, and must be compatible with fire-rated doors when used in such assemblies. They are often found in locations such as, but not limited to, commercial and industrial buildings, assemblies, banks, offices, hotels, high-rises, institutions, mercantile occupancies, etc.

Under the *SCA*, all stakeholders involved in the installation of maglocks have defined responsibilities to comply with the *SCA* and regulations pursuant to the *SCA*. Under the *SCA*, appropriate permits must be obtained for any undertaking for which a permit is required and is applicable to maglocks.

Maglock installations in new or existing buildings must be installed, tested, and verified in accordance with the applicable NBC(AE) provisions for safety, accessibility, and proper emergency egress. Article 2.2.3.3. Division C of the NBC(AE) requires fire alarm systems (FAS) plans to be submitted to the authority having jurisdiction (AHJ) for review. If the FAS is to be installed in a building described in Sentence 2.4.1.1.(3) Division C of the NBC(AE), it must be designed by a registered professional.

Article 3.2.4.5. of Division B in the NBC(AE) references standards for the installation and verification of a FAS: CAN/ULC-S524, “Standard for Installation of Fire Alarm Systems” and CAN/ULC-S537, “Standard for Verification of Fire Alarm Systems”. These standards define an ancillary device as a component connected to the FAS serving a life safety function but is not considered part of the FAS. Maglocks are considered ancillary devices and must be integrated with the building’s FAS. Article 3.2.9.1. of the NBC(AE), addresses the testing for the integration of such ancillary devices with the FAS and life safety systems.

Article 2.2.1.1. Division C of the NFC(AE) places responsibility on owners to ensure all fire protection and life safety systems within their buildings are properly maintained, inspected, and tested. These systems include but are not limited to, fire alarms, emergency lighting, sprinkler systems, smoke control systems, and maglocks, among others. These tasks must be performed by qualified individuals, in accordance with Subsection 2.2.4. Division C of the NFC(AE). Installation or modification of a life safety system requires testing to verify all components function together as intended. Article 2.2.1.2. Division C of the NFC(AE) requires inspection, testing, and maintenance activities be documented, with records retained on-site and made available to the AHJ upon request. Proper documentation and records support compliance with audit provisions for life safety systems designed to protect occupants during emergencies.

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Please note, when an owner or the owners authorized agent voluntarily installs a maglock (voluntarily means not a prescribed installation under the NBC(AE)) within a new or existing building, the installation, verification, testing and maintenance of the voluntarily installed maglock must comply with the *SCA*, NBC(AE) and NFC(AE). Both the NBC(AE) and NFC(AE) assume that all fire protection systems in a building, whether required by Code or voluntarily installed, will be designed in conformance with good fire protection engineering practice and will meet the appropriate installation requirements of the relevant codes and standards.

References

Safety Codes Act

Interpretation
1(1) **In this Act,**

f) "owner" includes a lessee, a person in charge, a person who has care and control and a person who holds out that the person has the powers and authority of ownership or who for the time being exercises the powers of authority of ownership:

(2) In this Act, a reference to "this Act" includes the regulations and bylaws made under this Act and any code, standards or body of rules declared to be in force pursuant to this Act.

Safety Codes Act

Application of Act
2(1) This Act applies to fire protection, barrier-free design and the design, manufacture, construction, installation, use, operation, occupancy and maintenance of

a) buildings,
b) electrical systems,
c) elevating devices,
d) gas systems,
e) plumbing systems,
f) pressure equipment, and
g) private sewage disposal systems.

Safety Codes Act

Responsibilities
Owners, care and control
5 The owner of any thing, process or activity to which this Act applies shall ensure that it meets the requirements of this Act, that the thing is maintained as required by the regulations and that when the process or activity is undertaken it is done in a safe manner.

Design duties
6 A person who creates, alters, has care and control of or owns a design or offers a design for use by others shall ensure that the design complies with this Act and that it is submitted for review or registered if required by this Act, and if the design is deregistered, the person shall provide

Manufacturers' duties
7 A person who manufactures any thing or undertakes a process or activity to which this Act applies shall ensure that the thing, the process or the activity complies with this Act.

Contractors' duties
8 A contractor who undertakes construction, operation or maintenance of or builds or installs any thing to which this Act applies shall ensure that this Act is complied with.

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Vendors' duties
9(1) A person who is a vendor in the ordinary course of business, other than as an employee or an agent, shall not advertise, display or offer for sale, for lease or for other disposal, or sell, lease or otherwise dispose of, any thing to which this Act applies unless that thing complies with this Act.

Professional services
11 A person permitted to affix stamps or seals pursuant to the Architects Act or the Engineering and Geoscience Professions Act shall ensure that any professional service the person renders to which this Act applies, including the affixing of stamps and seals, complies with this Act.

Safety Codes Act

Permits required
43(1) If this Act requires a person to have a permit to sell, construct, control or operate any thing or supervise, operate or undertake any process or activity, no person shall do so unless the person has the appropriate permit.

(2) If any thing to which this Act applies is approved by the regulations for a certain use or purpose, no person shall use that thing for any other use or purpose unless a safety codes officer issues a permit for that other use or purpose or it is an innocuous use or purpose.

(3) If the regulations require that any thing be approved before it is installed or operated, no person shall install or operate that thing unless a safety codes officer issues a permit for it.

(4) A permit under this Act does not authorize a person to do any thing, implement any process or engage in any activity that does not comply with any other enactment.

Code References

NBC(AE) and NFC(AE) Division A, Article 1.4.1.2.

1.4.1.2. Defined Terms
1) The words and terms in italics in this Code shall have the following meanings:
Authority having jurisdiction means a safety codes officer in the fire discipline exercising authority pursuant to designation of powers and terms of employment in accordance with the Safety Codes Act.

Building means any structure used or intended for supporting or sheltering any use or *occupancy*.

Constructor means a person who contracts with an owner or their authorized agent to undertake a *project* and includes an owner who contracts with more than one person for the *work* on a *project* or undertakes the *work* on a *project* or any part thereof.

Designer means the person responsible for the design.

Permit means, for the purposes of this Code, permission in writing to commence the use, *occupancy*, relocation, construction or demolition of any *building*.

Registered professional means an individual who qualifies as a
(a) *registered architectural professional*,
(b) *registered engineering professional*, or
(c) *licensed interior designer*.

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NBC(AE), Division B Article 3.2.4.5.

3.2.4.5. Installation and Verification of Fire Alarm Systems
1) Except as permitted by Article 3.2.4.19., fire alarm systems, including the voice communication capability where provided, shall be installed in conformance with CAN/ULC-S524, "Standard for Installation of Fire Alarm Systems."

2) Fire alarm systems shall be verified in conformance with CAN/ULC-S537, "Standard for Verification of Fire Alarm Systems," to ensure they are operating satisfactorily.

NBC(AE), Division B Article 3.2.9.1.

3.2.9. Integrated Fire Protection and Life Safety Systems
3.2.9.1. Testing
1) Where fire protection and life safety systems and systems with fire protection and life safety functions are integrated with each other, they shall be tested as a whole in accordance with CAN/ULC-S1001, "Standard for Integrated Systems Testing of Fire Protection and Life Safety Systems," to verify that they have been properly integrated. (See Note A-3.2.9.1.(1).)

A-3.2.9.1.(1) Testing of Fire Protection and Life Safety Systems. Building owners should verify that fire protection and life safety systems and their components (i.e. fire alarm systems, sprinklers, standpipes, smoke control, ventilation, pressurization, door hold-open devices, elevator recalls, smoke and fire shutters and dampers, emergency power, emergency lighting, fire pumps, generators, etc.), including their interconnections with other building systems, are functioning according to the intent of their design. CAN/ULC-S1001, "Standard for Integrated Systems Testing of Fire Protection and Life Safety Systems," provides the methodology for verifying and documenting that interconnections between building systems satisfy the intent of their design and that the systems function as intended by the Code.

Clause 6.1.5 of CAN/ULC-S1001 allows the Integrated Testing Coordinator to accept documented evidence of any tests that have been performed on a system as part of its acceptance testing for the purpose of demonstrating compliance with the integrated testing requirements of that standard, so as to avoid duplication of work.

NBC(AE), Division C Article 2.2.3.3.

2.2.3.3. Plans of Fire Alarm Systems
1) Before a fire alarm system is installed or altered, plans showing full details of the *building* in which it is to be installed shall be drawn to scale and submitted with the application to build.

2) Unless deemed by the *authority having jurisdiction* to be a minor *alteration* to an existing system, all plans for a fire alarm system referred to in Sentence 2.4.1.4.(1) shall be authenticated by a *registered professional*.

NBC(AE), Division C Sentence 2.4.1.4.(1)

2.4.1.4. Fire Alarm System Designs
1) If a fire alarm system is to be installed in a *building* described in Sentence 2.4.1.1.(3), the owner shall submit evidence to the *authority having jurisdiction*, before construction begins, that they have retained a *registered professional* to
a) design the system,
b) perform *field reviews* of the system during installation, and
c) witness verification of the system after installation.

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NBC(AE), Division C Sentence 2.4.1.1.(3)

2.4.1. Professional Involvement
2.4.1.1. General
3) For *buildings* other than those described in Sentence (1), the drawings and specifications shall be authenticated by one or more *registered professionals* qualified to engage in the appropriate combination of those areas of the practice of architecture and engineering that are applicable to the project.

CAN/ULC-S524, Standard for Installation of Fire Alarm Systems

1 Scope
1.1 This Standard describes the requirements for the *design* and installation of a *fire alarm system* with or without *voice communication* capability.
1.2 The installation of devices not covered by these requirements shall be in accordance with good engineering practice and manufacturer's published installation instructions and the following standards, as applicable:
c) CAN/ULC-S573, Installation of Ancillary Devices Connected to Fire Alarm Systems;
f) CAN/ULC-S1001, Integrated Systems Testing of Fire Protection and Life Safety Systems.

CAN/ULC-S524, Standard for Installation of Fire Alarm Systems

3 Glossary
3.9 *ANCILLARY DEVICE* – A device which has a life-safety application, is connected to the *fire alarm system*, but is not part of the *fire alarm system*.

Sentence 2.1.3.7(1)

2.1.3.7. Integrated Life Safety and Fire Protection Systems
1) Where life safety and fire protection systems are installed, they shall be tested in accordance with the NBC(AE).

Sentence 2.2.2.4.(1)

2.2.2.4. Inspection and Maintenance
1) Defects that interfere with the operation of *closures* in *fire separations* shall be corrected, and such *closures* shall be maintained to ensure that they are operable at all times by
a) keeping fusible links and other heat-actuated devices undamaged and free of paint and dirt,
b) keeping guides, bearings and stay rolls clean and lubricated,
c) making necessary adjustments and repairs to door hardware and accessories to ensure proper closing and latching, and
d) repairing or replacing inoperative parts of hold-open devices and automatic releasing devices.
(See Article 2.1.2.4. of Division C.)

Sentence 2.7.2.1.(4)

2.7.2.1. Exit Doors
4) When doors are equipped with electromagnetic locks, these locks shall be tested at intervals not greater than 12 months.

Sentence 2.7.2.2.(1)

2.7.2.2. Records
1) Records of tests required in Sentences 2.7.2.1.(2), (3) and (4) shall be retained in conformance with Article 2.2.1.2. of Division C.

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Sentence 6.1.1.2.(1)

6.1.1.2. Maintenance
1) Fire protection installations shall be maintained in operating condition.
(See Note A-6.1.1.2.(1).)

A-6.1.1.2.(1) Both the NBC(AE) and the NFC(AE) assume that all fire protection systems in a building, whether required by Code or voluntarily installed, will be designed in conformance with good fire protection engineering practice and will meet the appropriate installation requirements in relevant standards. Such good design is necessary to ensure that the level of safety established by the Code requirements is not reduced by a voluntary installation. Thus, a voluntarily installed system should be maintained in operating condition, at least to the extent that it was originally intended to function, in conformance with the applicable installation standards.

Articles 6.3.1.1. and 6.3.1.2.

Section 6.3. Fire Alarm and Voice Communication Systems
6.3.1. General (See Articles 2.2.4.3., 2.2.7.2. and 2.2.9.2. of Division C.)
6.3.1.1. Maintenance
1) Fire alarm and voice communication systems shall be maintained in operable condition at all times.

6.3.1.2. Inspection and Testing
1) Fire alarm systems shall be inspected and tested in conformance with CAN/ULC-S536, “Standard for Inspection and Testing of Fire Alarm Systems.”
2) Fire alarm and detection system components shall be accessible for purposes of inspection or maintenance.

Sentence 6.8.1.1.(1)

6.8.1.1. Testing and Maintenance
1) Interconnections between fire protection and life safety systems shall be tested and maintained in conformance with CAN/ULC-S1001, “Standard for Integrated Systems Testing of Fire Protection and Life Safety Systems.” (See Note A-6.8.1.1.(1).)

A-6.8.1.1.(1) Building owners must ensure that fire protection and life safety systems and their components (i.e. fire alarm systems, sprinklers, standpipes, smoke control, ventilation, pressurization, door hold-open devices, elevator recalls, smoke and fire shutters and dampers, emergency power, emergency lighting, fire pumps, generators, etc.), including their interconnections with other building systems, are functioning according to the intent of their design. CAN/ULC-S1001, “Standard for Integrated Systems Testing of Fire Protection and Life Safety Systems,” provides the methodology for verifying and documenting that interconnections between building systems satisfy the intent of their design and that the systems function as intended by the Code.

Clause 6.1.5 of CAN/ULC-S1001 allows the Integrated Testing Coordinator to accept documented evidence of any tests that have been performed on a system as part of its acceptance testing for the purpose of demonstrating compliance with the integrated testing requirements of that standard, so as to avoid duplication of work.

NFC(AE), Division C, Sentence 2.1.2.4.(1)

2.1.2.4. Exit Doors and Door Release Hardware
1) An *exit* door shall not be bolted, barred or locked other than with a device that complies with the NBC(AE). In addition, all door release hardware, latches and locks shall be maintained in good working condition at all times.

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NFC(AE), Division C, Sentence 2.2.1.1.(1)

2.2.1.1. Responsibility
1) Unless otherwise specified, the owner or the owner's authorized agent shall be responsible for carrying out the provisions of this Code.

NFC(AE), Division C, Sentences 2.2.1.1.(1), (8), and (9)

2.2.1.2. Records
1) Where this Code requires that audits be performed, plans be provided, or tests, inspections, maintenance or operational procedures be performed on a fire safety system, records shall be made and the original or a copy shall be retained at the premises for examination by the *authority having jurisdiction*.
8) Unless otherwise stated in Part 6, Division B, records indicating inspection, testing and maintenance of fire protection equipment shall be on appropriate forms in conformance with Sentence (9) and be retained in conformance with this Article.
9) Appropriate forms required by Sentence (8) and acceptable to the *authority having jurisdiction* are included in
a) CAN/ULC-S536, “Standard for Inspection and Testing of Fire Alarm Systems,”
b) CAN/ULC-S537, “Standard for the Verification of Fire Alarm Systems,” or
c) CAN/ULC-S1001, “Standard for Integrated Systems Testing of Fire Protection and Life Safety Systems.”

NFC(AE), Division C, Sentence 2.2.4.3.(2)

2.2.4. Qualifications
2.2.4.3. Fire Alarm and Voice Communication Systems
2) Persons are considered qualified to make operational, inspect, test and maintain fire alarm and voice communication systems and sign applicable documentation when they have acquired
a) a certificate of training in this area of study from a public post-secondary educational institution, including
i) a journeyperson's electrician certificate on or after September 1, 1991, or
ii) a journeyperson's electrician certificate prior to September 1, 1991, and a fire alarm course recognized by the Provincial Fire Administrator,
b) Fire Alarm Technician certification issued by the Canadian Fire Alarm Association (CFAA) and that certification shall be maintained as per the CFAA's certification policy.

NFC(AE), Division C, Sentence 2.2.4.8.(1)

2.2.4. Qualifications
2.2.4.8. Integrated Fire Protection and Life Safety Systems
1) A person qualified to be an integrated testing coordinator shall be a registered engineering professional knowledgeable and experienced in the design, installation and operation of fire protection and life safety systems and in the fire protection and life safety functions of *building* systems. (See Note A-2.2.4.8.(1).)

A-2.2.4.8.(1) A non-registered engineering professional may do the groundwork for the integrated systems testing, provided that the registered engineering professional supervises, approves and officially signs off on that work.

NFC(AE) Division C, Sentences 2.2.7.2.(1), (2), (3) and (4)

2.2.7.2. “NFC(AE)” Fire Protection Equipment
1) In addition to the requirements of Sentence 6.1.1.3.(1) of Division B, before repairs or alterations are made to fire protection installations, including but not limited to fire extinguishing systems and fire alarm and detection systems, a procedure of notification acceptable to the fire department shall be established, and the procedure may include the notification of the fire department and the building occupants. (See Note A-6.1.1.3.(1) of Division B.)
2) If a fire alarm and detection system, or part thereof, is inoperative for more than 2 h for any reason, the owner shall notify the fire department and, when directed, provide acceptable surveillance within the building continuously until the fire alarm and detection system is restored to operating condition.

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3) Procedures acceptable to the fire department shall be developed to notify occupants if a fire or other emergency occurs while the fire alarm and detection system is inoperative. (See Note A-2.2.7.2.(3).)
4) If a person who carries out the maintenance of fire protection installations, including but not limited to portable fire extinguishers, fire extinguishing systems, fire alarm systems or fire alarm devices, discovers that a device or system is inoperative or defective and the owner or their authorized agent is unwilling or unable to correct the defect, the person carrying out the maintenance shall forthwith notify, in writing, the fire department and the owner or their authorized agent of that notification.

NFC(AE) Division C, Sentence 2.2.9.2.(1)

2.2.9.2. Fire Alarm and Voice Communication System Audits
1) When an existing fire alarm system has not been subjected to an initial verification as required by CAN/ULC-S524, "Standard for Installation of Fire Alarm Systems," and described in CAN/ULC-S537, "Standard for the Verification of Fire Alarm Systems," the *authority having jurisdiction* may require the system to be audited in conformance with CAN/ULC-S536, "Standard for Inspection and Testing of Fire Alarm Systems," unless the *authority having jurisdiction* is satisfied that the system has been maintained and is functioning in a manner acceptable to the *authority having jurisdiction*.

CSA C22.1, Canadian Electrical Code, Part 1

32-000 Scope (see Appendix B)

1) This Section applies to the installation of electrical local fire alarm systems, permanently connected smoke alarms, carbon monoxide alarms, and fire pumps required by the National Building Code of Canada.

Appendix B

Rule 32-000

For further information pertaining to the installation of fire alarm systems, reference should be made to CAN/ULC-S524.

It is essential that fire alarm systems be maintained in an operating condition at all times. The inspection, maintenance, and testing procedures are detailed in the National Fire Code of Canada or the appropriate provincial fire code.

Application

This interpretation applies to the installation, maintenance, testing, and verification of electromagnetic locks in both new and existing buildings.

Interpretation

This interpretation clarifies the requirements for installation, verification, inspection, testing, and maintenance for maglocks installed in a building connected to a FAS in new or existing buildings. Only individuals qualified under the NFC(AE) may perform these activities.

1. Permits are required for installing maglocks, whether modifying an existing system or installing a new system.
2. Where the NBC(AE) requires professional involvement due to a building's size, occupancy, or complexity, a registered engineering professional must be retained to ensure compliance.

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3. Installation of maglocks in a building must comply with the NBC(AE).

   - Maglocks are recognized as an ancillary device requiring connection to the FAS. They must be:
     - installed in accordance with the with CAN/ULC-S524, “Standard for Installation of Fire Alarm Systems.”
     - verified in conformance with CAN/ULC-S537, “Standard for Verification of Fire Alarm Systems.”
     - integrated testing must be completed in accordance with CAN/ULC-S1001, “Standard for Integrated Systems Testing of Fire Protection and Life Safety Systems”.

4. Life safety systems must comply with the NFC(AE) throughout the building’s lifecycle.

   - Maglocks forming part of these systems must be:
     - inspected and tested in conformance with CAN/ULC-S536, “Standard for Inspection and Testing of Fire Alarm Systems.”
     - integrated testing must be tested and maintained in accordance with CAN/ULC-S1001, “Standard for Integrated Systems Testing of Fire Protection and Life Safety Systems”.

5. Records of inspection, testing, and maintenance must be:

   - retained on-site.
   - made available to the AHJ upon request; and maintained in a format acceptable to the AHJ, as outlined in:
     - CAN/ULC-S536, “Standard for Inspection and Testing of Fire Alarm Systems,”
     - CAN/ULC-S537, “Standard for the Verification of Fire Alarm Systems,” or
     - CAN/ULC-S1001, “Standard for Integrated Systems Testing of Fire Protection and Life Safety Systems.”

This INTERPRETATION is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Fire Code – 2023 Alberta Edition

Issued by the Provincial Fire and Building Administrators

[Original Signed]
Tina Parker

[Original Signed]
Paul Chang

Alberta Municipal Affairs – Technical and Corporate Services
Email: safety.services@gov.ab.ca Phone: 1-866-421-6929

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